



Vinay Joshi

Microsoft
Generative AI
reasoning
code generation

	All	Since 2021
Citations	1194	1119
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	1 article	4 articles
	not available	available

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TITLE	CITED BY	YEAR
Geak: Introducing triton kernel ai agent & evaluation benchmarks J Wang, V Joshi, S Majumder, X Chao, B Ding, Z Liu, PP Brahma, D Li, ... arXiv preprint arXiv:2507.23194	6	2025
Apparatuses, methods, and systems for instructions for matrix transpose K Pillai, CJ Hughes, V Joshi, OJ Omer US Patent App. 18/401,410		2025
TaDA: Training-free recipe for Decoding with Adaptive KV Cache Compression and Mean-centering V Joshi, PP Brahma, Z Liu, E Barsoum arXiv preprint arXiv:2506.04642		2025
Systems, method, and apparatus for quality and capacity-aware grouped query attention V Joshi, OJ Omer, P Laddha, S Sinha US Patent App. 18/900,006		2025
Gated unit for a gated recurrent neural network M Le Gallo-Bourdeau, VM Joshi, A Sebastian, M Stanisavljevic US Patent 12,093,802	1	2024
QCQA: Quality and capacity-aware grouped query attention V Joshi, P Laddha, S Sinha, OJ Omer, S Subramoney arXiv preprint arXiv:2406.10247	7	2024
CiMNet: Towards Joint Optimization for DNN Architecture and Configuration for Compute-In-Memory Hardware S Kundu, A Sarah, V Joshi, OJ Omer, S Subramoney arXiv preprint arXiv:2402.11780	1	2024

TITLE	CITED BY	YEAR
A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference M Le Gallo, R Khaddam-Aljameh, M Stanisavljevic, A Vasilopoulos, ... Nature Electronics 6 (9), 680-693	297	2023
Conductance drift corrections in neuromorphic systems based on crossbar array structures VM Joshi, S Haefeli, M Le Gallo-Bourdeau, IB Kara, A Sebastian US Patent 11,663,458	3	2023
Electronic system for computing items of an outer product matrix VM Joshi, A Sebastian, M Le Gallo-Bourdeau, IB Kara, C Piveteau US Patent 11,436,302	2	2022
Training of artificial neural networks C Piveteau, A Sebastian, M Le Gallo-Bourdeau, VM Joshi US Patent 11,373,092	12	2022
Hybrid in-memory computing architecture for the training of deep neural networks V Joshi, W He, J Seo, B Rajendran 2021 IEEE International Symposium on Circuits and Systems (ISCAS), 1-5	1	2021
Accurate deep neural network inference using computational phase-change memory V Joshi, ML Gallo, I Boybat, S Haefeli, C Piveteau, M Dazzi, B Rajendran, ... Nature Communications	577	2020
Mixed-precision deep learning based on computational memory SR Nandakumar, ML Gallo, C Piveteau, V Joshi, G Mariani, I Boybat, ... Frontiers of Neuroscience	143	2020
ESSOP: Efficient and Scalable Stochastic Outer Product Architecture for Deep Learning V Joshi, G Karunaratne, ML Gallo, I Boybat, C Piveteau, A Sebastian, ... International Symposium on Circuits and Systems 2020	3	2020
Phase-Change Memory Models for Deep Learning Training and Inference S. R. Nandakumar, Irem Boybat, Vinay Joshi, Chritophe Piveteau, Manuel Le ... 26th IEEE International Conference on Electronics Circuits and Systems	37 *	2019

TITLE	CITED BY	YEAR
Deep learning acceleration based on in-memory computing E Eleftheriou, M Le Gallo, SR Nandakumar, C Piveteau, I Boybat, V Joshi, ... IBM Journal of Research and Development 63 (6), 7: 1-7: 16	40	2019
Computational memory-based inference and training of deep neural networks A Sebastian, I Boybat, M Dazzi, I Giannopoulos, V Jonnalagadda, V Joshi, ... 2019 Symposium on VLSI Technology, T168-T169	52	2019
High accuracy sensor fault detection for energy management applications V Joshi, O Desai, A Kowli 2017 IEEE International Conference on Signal Processing, Informatics ...	12	2017
Deep learning acceleration based on in-memory computing C Piveteau, I Boybat, V Joshi, R Khaddam-Aljameh, M Dazzi, ...		